# Guanhao Wang

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Computer science undergraduate focused on game development, environmental storytelling, and physical interaction.

## EDUCATION

**University of Wisconsin–Madison** June 2025 to present
*Computer Science Undergraduate · Expected graduation May 2027 · GPA 3.724/4.000*

**The Pennsylvania State University** August 2023 to June 2025
*Undergraduate Study in Computer Science · GPA 3.94/4.00 · Transferred to UW–Madison*

## GAMES AND INTERACTIVE PROJECTS

**Coal Echoes** ↗ 2026
*Narrative Exploration Game · In development · Unity 6, HDRP, C#, Blender*

- Designed a four level first person exploration game about 1970s working class memories at an abandoned coal plant; built photography, album, scrapbook, and inspection systems for object discovery through the camera.
- Used an AI agent through Blender MCP to assist with parametric modeling and texture baking for industrial assets, including cableway structures and a guardhouse, then integrated them into Unity.
- Generated main menu newspaper photo materials at runtime from live HDRP Custom Pass camera captures, using HLSL halftone logic adapted from Daniel Ilett (MIT).

**Project Fireflies** ↗ 2026
*Interactive Light Installation · Completed · ESP32, Arduino, Unity, UDP*

- Programmed 420 LEDs across three swarms, using radar and light sensors to translate visitor movement and human light sources into firefly disturbance and synchronization.
- Implemented two independently responsive zones, one ambient swarm, and a UDP interface for Unity synchronization commands.

**Seafloor Revival** ↗ July to August 2025
*GMTK Game Jam 2025 · UI Development and Art Integration · Unity, C#*

- Built all game UI and related C# behavior, integrated interface art in a six person team, and contributed to a released browser prototype of a 2.5D underwater building game.

**Mooncake** ↗ February to April 2025
*Interactive Graphics · Three.js, JavaScript*

- Built a Three.js mooncake scene for the Mid Autumn Festival with STL models, shadowed spotlighting, and drifting star particles; added live controls for light, camera orbit, and particles, documented in a technical report.

## PROFESSIONAL EXPERIENCE

**IQVIA** ↗ July to August 2024
*Technology Solutions Intern · Shanghai, China*

- Wrote and maintained 100+ pages of user guides for a pharmaceutical WeChat Mini Program upgrade, covering two user types, four scenarios, and three device contexts.
- Performed user acceptance testing and reported 10+ defects in Jira with reproduction steps; organized 1,000+ data tags into a three level hierarchy for user filtering.

## PUBLICATION

**Human Computer Interaction in Indie Games** ↗ 2023
*Sole Author*

- Analyzed implicit guidance, environmental storytelling, and player attachment in FAR Lone Sails; published in Lecture Notes in Education Psychology and Public Media, vol. 18, pp. 194 to 199.

## SKILLS AND LANGUAGES

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|---|---|
| **Games and interaction** | Unity, C#, HDRP, uGUI, Shader Graph, Blender, ESP32, Arduino |
| **Programming and tools** | Python, Java, C, JavaScript, Three.js, Git, Jira |
| **Languages** | Mandarin native, English fluent, Japanese basic |